

Experimental vapour–liquid equilibrium data of the quaternary system Methanol (1) + Isopropyl alcohol (2) + Water (3) + Glycerol (4) along with Isopropyl alcohol (2) + Glycerol (4) and Isopropyl alcohol (2) + Water (3) binary data at atmospheric and sub-atmospheric pressures

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ARTICLE INFO

Article history:

Received 24 June 2015

Received in revised form

26 September 2015

Accepted 26 September 2015

Available online 30 September 2015

Keywords:

Quaternary vapor–liquid equilibrium

Methanol

Isopropyl alcohol

Water

Glycerol

ABSTRACT

The vapour–liquid equilibrium (VLE) behaviour of the quaternary system, Methanol (1) + Isopropyl alcohol (IPA) (2) + Water (3) + Glycerol (4) at 53.33, 66.66, 79.99 and 95.2 kPa, has been studied experimentally using a Sweitoslawsky-type ebulliometer. The VLE data of this quaternary system at sub-atmospheric and ambient conditions will be useful in the process intensification design of bio-diesel production. Experimental VLE data were compared with predictions from Wilson and NRTL models using only the constituent binary interaction parameters. The predictions of equilibrium temperatures thus obtained using Wilson model compared well with the experimental values with an average root mean square deviation (rmsd T) of 0.3153 K. The binary system IPA (2) + Glycerol (4) was studied experimentally at 53.33, 66.66, 79.99 and 94.93 kPa and the data correlated with Wilson, NRTL, UNIQUAC and Peng–Robinson (PR) and Soave–Redlich–Kwong (SRK) equations of state (EoS) combined with Wong–Sandler (WS) mixing rules. Experimental data of this binary system was also compared with predictions from UNIFAC model. The VLE data of the binary system IPA (2) + Water (3) at 94.66 kPa is reported here and correlated with Wilson and NRTL activity-coefficient models. Wilson model represented the binary systems better than other models with a mean rmsd T of 0.2853 K for IPA (2) + Glycerol (4) and an rmsd T of 0.1390 for the IPA (2) + Water (3) system. MATLAB programming has been used for all the calculations.

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1. Introduction

The experimental phase behaviour study of the quaternary system reported here, Methanol (1) + Isopropyl alcohol (IPA) (2) + Water (3) + Glycerol (4) is in continuation with our previous studies on systems found in bio-diesel processing [1,2]. The binary systems, IPA (2) + Glycerol (4) and IPA (2) + Water (3) are constituent systems of this quaternary system and are also components of common industrial streams. IPA was chosen for this study

as it is a potential key component in the production of high-value added products such as Isopropyl myristate, Isopropyl palmitate, Isopropyl oleate, linoleate etc as part of bio-diesel processing. This can be achieved through enzymatic transesterification which is a sustainable and economical green process.

The other four binary sub-systems of this quaternary system have been reported previously by the authors at atmospheric and sub-atmospheric pressures [1,3]. The Methanol (1) + Water (3), Methanol (1) + Glycerol (4) and Water (3) + Glycerol (4) systems [1] were studied at (15.19, 29.38, 42.66, 56.03, 67.38 and 95.3) kPa, (32.02 and 45.03) kPa and (14.19, 29.38, 41.54, 54.72, 63.84 and 95.3) kPa respectively. The Wilson model was found to represent these three binary systems very well. The Methanol (1) + IPA (2) binary system was studied at (53.3, 66.7, 79.9 and 94.79) kPa and was represented best by the NRTL model [3]. Oliveira et al. [4] have

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reported the vapour–liquid equilibrium (VLE) data of IPA (2) + Glycerol (4) system at atmospheric pressure as part of their study of systems containing glycerol and which are part of production and down-stream processing of bio-diesel. Experimental VLE data of the IPA (2) + Water (3) binary system has been reported by many authors. In the present study, the authors found that the best-fitting model for this binary system predicts an azeotrope at the pressure studied. This is in agreement with data reported in literature [5–8]. The authors did not find any experimental data in literature for the quaternary system at the pressures reported in this work.

2. Experimental

2.1. Materials used

Methanol (>0.998 mol fraction, HPLC grade) provided by SD Fine Chemicals, India, IPA (>0.998 mol fraction, HPLC grade) supplied by Sigma Aldrich, glycerol (>0.995 mol fraction) provided by SD Fine Chemicals India and double-distilled water were used for experimental studies. The chemicals were passed through molecular sieves to remove any trace amounts of water and stored in dessicators prior to experimentation to prevent absorption of moisture and were used without any further purification. The densities and refractive indices of the pure components measured at 293.15 and 298.15 K are compared with literature values [9–12] in Table 1.

2.2. Apparatus

A Sweitoslawsky-type ebulliometer, similar to the one described by Hala et al. [13], has been used for experimentation. The details of the construction of the ebulliometer and the experimental procedure followed are described by Vittal Prasad et al. [14]. A schematic diagram of the ebulliometer is given in Fig. 1. The ebulliometer is connected to a vacuum pump and a nitrogen gas cylinder, along with a closed end U-tube mercury manometer in line to maintain the pressure to within 0.067 kPa of the desired value. This is done by manually adjusting the opening of the needle valve of the gas cylinder or by opening the bypass line of the vacuum pump. A WIKA digital thermometer (Model CTH6200, ± 0.05 K) calibrated by point-to-point comparison with a WIKA hand-held calibrator (Model CEP3000) is used for measuring the equilibrium

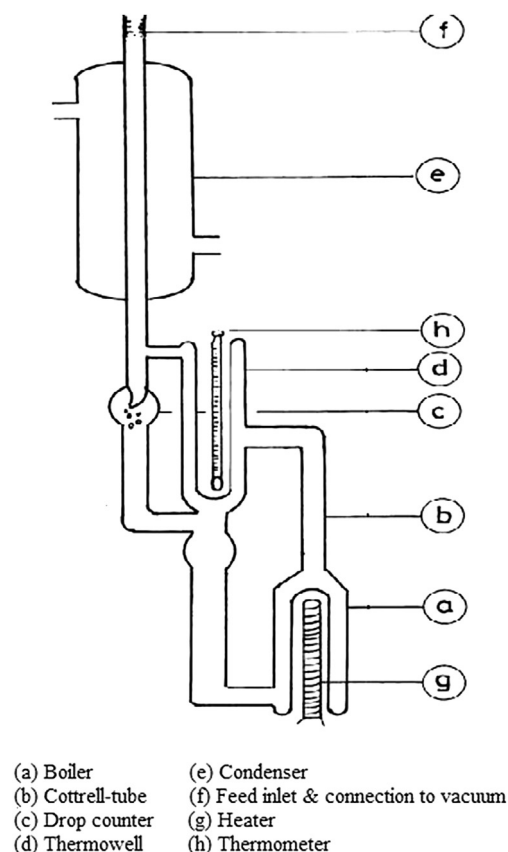


Fig. 1. Schematic diagram of Sweitoslawski-type ebulliometer.

temperatures with an accuracy of ± 0.01 K. The thermometer is placed in the thermo-well containing glycerol to note the equilibrium temperature for the (vapour + liquid) mixture impinging on the thermo-well from the Cottrell tube of the ebulliometer. The equilibrium temperature is recorded after steady-state conditions, judged by the constancy of temperature and uniformity of the drop rate maintained for at least 1800 s. The mixtures to be studied were prepared gravimetrically by weighing the required amounts of the pure liquids and stirring them well to make homogeneous mixtures

Table 1
Comparison of density, ρ and refractive index, n_D data with literature data.

Chemical	CAS no.	Source	Mole fraction purity	T (K)	ρ ($\times 10^3$ kg m $^{-3}$)		n_D	
					This work	Reference	This work	Reference
Methanol	67-56-1	SDFCL	>0.998	293.15	0.79162	0.79104 ^a	1.32864	1.3284 ^a
				298.15	0.78662	0.78656 ^b 0.78637 ^a	1.32666	1.3267 ^b 1.32652 ^a
IPA	67-63-0	Sigma Aldrich	>0.998	293.15	0.78691	0.78545 ^a	1.37736	1.3772 ^a
				298.15	0.78245	0.7827 ^c 0.78126 ^a 0.78130 ^d	1.37522	1.3754 ^c 1.3752 ^a 1.3750 ^d
Water				293.15	0.99821	0.998206 ^a	1.33305	1.332988 ^a
				298.15	0.99715	0.99705 ^a	1.33275	1.3325 ^a
Glycerol	56-81-5	SDFCL	>0.995	293.15	1.26135	1.26134 ^a	1.4744	1.4746 ^a
				298.15	1.25613	1.2559 ^a	1.47385	1.473 ^a

Standard uncertainty $u(\rho) = \pm 0.1$ kg m $^{-3}$, expanded uncertainty (95% confidence level) $u(n_D) = \pm 5 \cdot 10^{-5}$.

^a Reference [9].

^b Reference [10].

^c Reference [11].

^d Reference [12].

Table 2
Pure component vapour pressures.

Chemical	P_0 (kPa)	T (K)	Reference
IPA	52.97	340	[17]
	53.33	340.05	This work
	66.7	345.25	This work
	79.9	349.5	This work
	94.93	354.5	This work
	101.3	355.35	[12]
	101.3	355.41	[18]
	101.32	355.37	[19]
Water	101.32	355.392	[10]
	94.66	371.1	This work
	100.9	373.04	[20]
Glycerol	101.32	373.16	[21]
	94.93	560.68	This work
	101.32	563.2	[10]

Table 3
Experimental VLE data of IPA (2) + Glycerol (4) binary system at different pressures.

x_2^a	T (K)	y_2^b	x_2^a	T (K)	y_2^b
$P = 53.33$ kPa			$P = 66.66$ kPa		
0	539.13	0	0	547.20	0
0.0839	341.00	0.999984	0.0239	346.65	0.999978
0.1685	340.50	0.999986	0.0719	345.68	0.999981
0.1928	340.35	0.999986	0.0839	345.36	0.999981
0.3028	340.10	0.999986	0.1685	345.15	0.999981
0.3644	339.95	0.999986	0.2538	345.05	0.999981
0.4015	339.90	0.999986	0.3644	344.90	0.999981
0.4512	339.80	0.999986	0.5011	344.70	0.999981
0.5011	339.68	0.999986	0.5512	344.65	0.999981
0.5512	339.60	0.999986	0.6522	344.60	0.999981
0.6522	339.55	0.999986	0.7455	344.57	0.999981
0.7455	339.45	0.999986	0.8441	344.50	0.999981
0.8441	339.25	0.999986	0.8958	344.62	0.999982
0.8958	339.35	0.999987	0.9218	344.65	0.999982
0.9218	339.46	0.999987	0.9478	344.70	0.999982
0.9478	339.56	0.999987	0.9638	344.75	0.999984
0.9638	339.52	0.999988	0.9739	344.85	0.999988
0.9739	339.89	0.999991	1	345.25	1
1	340.05	1			
$P = 79.99$ kPa			$P = 94.93$ kPa		
0	554.05	0	0	560.68	0
0.0839	351.02	0.999971	0.0839	354.84	0.999963
0.0948	349.75	0.999972	0.1685	354.09	0.999967
0.1685	349.45	0.999974	0.2538	353.80	0.999967
0.2538	349.25	0.999975	0.3644	353.65	0.999968
0.3644	349.09	0.999975	0.4512	353.45	0.999968
0.4512	348.97	0.999975	0.5011	353.30	0.999968
0.5011	348.91	0.999975	0.5512	353.15	0.999968
0.5512	348.87	0.999975	0.5890	353.05	0.999968
0.6522	348.83	0.999975	0.6522	353.19	0.999968
0.7455	348.80	0.999975	0.7031	353.30	0.999968
0.8441	348.83	0.999975	0.7455	353.35	0.999968
0.8958	348.93	0.999975	0.8441	353.45	0.999968
0.9218	349.01	0.999976	0.8958	353.52	0.999968
0.9478	349.06	0.999977	0.9218	353.52	0.999968
0.9739	349.15	0.999985	0.9478	353.61	0.999968
1	349.50	1	0.9739	353.75	0.999971
			1	354.50	1

Standard uncertainties u are $u(P) = 0.17$ kPa, $u(T) = 0.01$ K, $u(x_i) = 10^{-5}$.^a x liquid phase mole fraction.^b y vapour phase mole fraction (calculated using Wilson model).

before charging the mixtures into the still. A Shimadzu balance (Model No. AUX 220), capable of recording weights up to ± 0.0001 g, was used for this purpose. After the mixture is charged in to the ebulliometer, vacuum is applied gradually and then the heating rate is slowly increased and adjusted to produce the required boil-up rate so that a drop count of about 30 drops per minute is

achieved. This is in accordance with the suggestion of Hala et al. [13]. The VLE experiment is first done at the lowest pressure. After the measurement of the equilibrium temperature at this pressure, vacuum is released and the pressure is increased to the next higher value to be studied. The binary system IPA (2) + Glycerol (4) and the quaternary system, Methanol (1) + IPA (2) + Water (3) + Glycerol (4) have been studied at 53.33, 66.66, 79.99 kPa and at 94.93 and 95.2 kPa respectively. The IPA (2) + Water (3) binary system has been studied at 94.66 kPa. The densities and refractive indices of the pure components were measured with a Rudolph Research Analytical automatic densitometer (Model DDM 2911) and Schmidt + Haensch Refractometer (Model DSR λ) respectively. The measurement technique used in the densitometer is mechanical oscillator method.

3. Results and discussion

3.1. Pure component vapour pressure

The Antoine constants used in the present study have been found to represent the pure component vapour pressures well over the entire pressure range of interest in our experimentation. In a previous study by the authors [15], the mean deviation between experimental vapour pressures and vapour pressures calculated using the Antoine constants from Reid et al. [16] was found to be less than 0.5% for Methanol and IPA. Pure component vapour pressures of methanol, IPA and Glycerol have been compared with values from literature. The results presented in Table 2 show that the experimental and literature values are in good agreement.

3.2. Isobaric binary VLE

The experimental T - x_1 data of IPA (2) + Glycerol (4) binary system at (53.3, 66.7, 79.9 and 94.93) kPa is reported in Table 3 while the data of IPA (2) + Water (3) system at 94.66 kPa is reported in Table 4. The experimental VLE data of IPA (2) + Glycerol (4) has been fitted to the Wilson, NRTL, UNIQUAC activity coefficient models and Peng–Robinson (PR), Soave–Redlich–Kwong (SRK) Equations of State (EoS). The EoS were combined with the Wong–Sandler (WS) mixing rules. Experimental data of IPA (2) + Glycerol (4) was also compared with the predictions of the group contribution method UNIFAC. The pure component

Table 4
Experimental VLE data of IPA (2) + Water (3) at 94.66 kPa.

x_2^a	T (K)	y_2^b
0	371.10	0
0.0100	367.60	0.1370
0.0600	359.40	0.3802
0.1010	357.60	0.4435
0.1550	356.20	0.4872
0.2340	355.10	0.5249
0.3500	354.00	0.5657
0.4000	353.70	0.5826
0.4500	353.30	0.6001
0.5070	353.00	0.6216
0.5940	352.65	0.6583
0.6770	352.40	0.6997
0.7700	352.40	0.7584
0.8300	352.40	0.8045
0.9000	352.80	0.8708
0.9500	353.10	0.9292
1	354.05	1

Standard uncertainties u are $u(P) = 0.17$ kPa, $u(T) = 0.01$ K, $u(x_i) = 10^{-5}$.^a x liquid phase mole fraction.^b y vapour phase mole fraction (calculated using Wilson model).

parameters such as critical temperature, critical pressure, accentric factor etc. and model equations for Wilson, NRTL, UNIQUAC and UNIFAC are from Reid et al. [22]. The assignment of functional groups was done according to group specification given in Reid et al. [22]. The PR, SRK EoS and WS mixing rules equations as given by Orbey and Sandler [23] were used. The IPA (2) + Water (3) VLE data has been correlated with the Wilson and NRTL models. The pure component data used in the correlation of experimental data with the various models is given in Table 5. The Antoine constants from Reid et al. [16] have been used for vapour pressure computation in this work. $T_{\max}$ and $T_{\min}$ indicate the range of temperature for which the given Antoine constants are valid. It can be observed that the Antoine constants used are valid for the entire range of T of this study for all the components.

The total pressure equation is used in the calculations for binary data reduction by Barker's method.

$$P_{\text{cal}}/kPa = (P_1^0/kPa)x_1\gamma_1F_1 + (P_2^0/kPa)x_2\gamma_2F_2 \quad (1)$$

Where, P_i^0 is the vapour pressure of the pure components, x_i are the liquid-phase mole fractions while γ_i are the activity coefficients. The vapour phase non-ideality correction factor, F_i in Eq. (1) is close to unity for low and ambient pressures according to Reid et al. [18]. Hence Eq. (1) is reduced to,

$$P_{\text{cal}}/kPa = (P_1^0/kPa)x_1\gamma_1 + (P_2^0/kPa)x_2\gamma_2 \quad (2)$$

The non-linear regression analysis methods, Powell's Dog-Leg and Levenberg–Marquardt methods, as described by Madsen et al. [24] were the numerical methods used in correlating experimental data to find optimum parameters of different models used, by minimizing the difference between calculated and experimental pressure. The objective function used is,

$$\Phi = \sum \left(\frac{P_{\text{cal}}}{P_{\text{exp}}} - 1 \right)^2 \quad (3)$$

Powell's Dog-Leg method was found to be superior in all the calculations. Hence the results from Dog-Leg method only have been presented here. MATLAB programmes employing the chosen mathematical and thermodynamic models for reduction of experimental data have been used for the calculations.

The results of the correlation of experimental data with the models are presented in Tables 6 and 7 for IPA (2) + Glycerol (4) and IPA (2) + Water (3) binaries respectively. The deviation of temperatures obtained from the correlation of experimental data with the activity coefficient models, EoS and predictions using the group contribution method UNIFAC, from the experimentally measured values is expressed in terms of root mean square deviation (rmsd), represented by $\sigma(T)$. These values are presented along with the model parameters in Tables 6 and 7 rmsd has been defined in the footnotes of these Tables. Optimised parameters of the NRTL activity coefficient model were used in the EoS model calculations for the IPA (2) + Glycerol (4) binary system.

Table 5

Pure component data.

Chemical	T_c (K) ^a	P_c (kPa) ^a	ω ^a	A^b	B^b	C^b	$T_{\max}^b$ (K)	$T_{\min}^b$ (K)
Methanol	512.6	8090	0.556	18.5875	3626.55	−34.29	364	257
IPA	508.3	4760	0.665	18.6929	3640.2	−53.54	374	273
Water	647.3	22,120	0.344	18.3036	3816.44	−46.13	441	284
Glycerol	726	6680	0.512 ^c	17.2392	4487.04	−140.2	600	440

^a Critical temperature (T_c), critical pressure (P_c), accentric factor (ω). Reference [22].

^b Antoine constants $\ln P/\text{Torr} = A - B/(T/K + C)$; 1 Torr = 0.133 kPa. Reference [16].

^c ω from Reference [16].

Table 6

Representation of VLE measurements by the different models for IPA (2) + Glycerol (4).

Parameter ^a	Wilson	NRTL	UNIQUAC	UNIFAC	PR + WS	SRK + WS
$P = 53.3$ kPa						
A_{12}	−0.0252	0.7893	−17.512	—	0.001	1
A_{21}	−0.0144	0.9075	316.9339	—	—	—
α	—	−1.57	—	—	—	—
$\sigma(T)^b$	0.2978	0.6108	2.2313	20.4058	0.3404	24.8165
$P = 66.7$ kPa						
A_{12}	−0.0071	0.7683	−126.111	—	0.05	—
A_{21}	−0.0144	0.8549	551.3246	—	—	—
α	—	−1.9217	—	—	—	—
$\sigma(T)^b$	0.2897	1.1939	4.3718	30.0685	1.3748	—
$P = 79.9$ kPa						
A_{12}	−0.0250	0.8017	−17.1464	—	0.0309	1
A_{21}	−0.0150	0.9275	335.2896	—	—	—
α	—	−1.4709	—	—	—	—
$\sigma(T)^b$	0.3052	0.6024	2.339	25.8636	0.3661	34.9595
$P = 94.93$ kPa						
A_{12}	−0.0232	0.8124	−2.1524	—	0.038	1
A_{21}	−0.0095	0.9167	311.3495	—	—	—
α	—	−1.4306	—	—	—	—
$\sigma(T)^b$	0.2486	0.4323	2.3953	21.5607	0.4301	27.35

^a Wilson: $A_{12} = \Lambda_{12}$, $A_{21} = \Lambda_{21}$; NRTL: $A_{12} = \tau_{12}$, $A_{21} = \tau_{21}$; UNIQUAC:

$A_{12} = (u_{12} - u_{22})/R$, $A_{21} = (u_{21} - u_{11})/R$; PR/SRK + WS: $A_{12} = k_{12} = k_{21}$.

^b $\sigma(T) = [\sum (T_{\text{exp}} - T_{\text{cal}})^2/n]^{0.5}$ K, where n is the number of observations.

Table 7

Representation of VLE measurements by the different models for IPA (2) + Water (3).

Parameter ^a	Wilson	NRTL
A_{12}	0.1362	0.6933
A_{21}	0.8020	1.6785
α	—	0.6806
$\sigma(T)^b$	0.1390	0.2790

^a Wilson: $A_{12} = \Lambda_{12}$, $A_{21} = \Lambda_{21}$; NRTL: $A_{12} = \tau_{12}$, $A_{21} = \tau_{21}$.

^b $\sigma(T) = [\sum (T_{\text{exp}} - T_{\text{cal}})^2/n]^{0.5}$ K, where n is the number of observations.

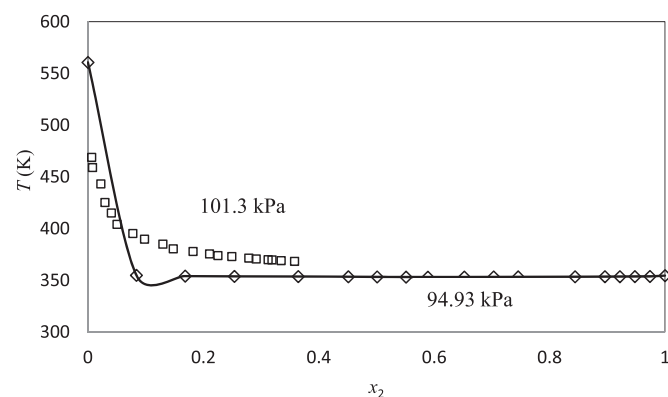


Fig. 2. Experimental T - x_2 data of IPA (2) + Glycerol (4). ($\diamond$) This work $T_{\text{exp}}-x_2$ exp at 94.93 kPa; (solid line) this work $T_{\text{cal}} - x_2$ cal from Wilson model parameters at 94.93 kPa; ($\square$) reference [4].

Among the different models used for correlating experimental data of the IPA (2) + Glycerol (4), the binary system is best represented by the Wilson model, followed closely by the PR + WS EoS, although the other activity coefficient models and PR EoS with WS mixing rules also represent this system reasonably well. Oliveira et al. [4] have reported VLE data of this binary system in the

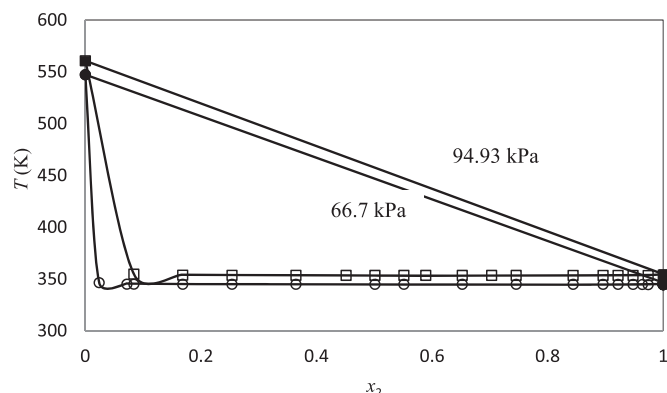


Fig. 3. Effect of pressure on T - x - y behaviour of IPA(2) + Glycerol(4) system. (○) $T_{\text{exp-}x_2}$ at 66.7 kPa; (●) $T_{\text{exp-}y_2}$ at 66.7 kPa; (□) $T_{\text{exp-}x_2}$ at 94.93 kPa; (■) $T_{\text{exp-}y_2}$ at 94.93 kPa; (solid line) $T_{\text{cal-}x_2}$ and $T_{\text{cal-}y_2}$ at 66.7 and 94.93 kPa; $T_{\text{cal-}x_2}$ and y_2 values have been calculated using optimized Wilson model parameters.

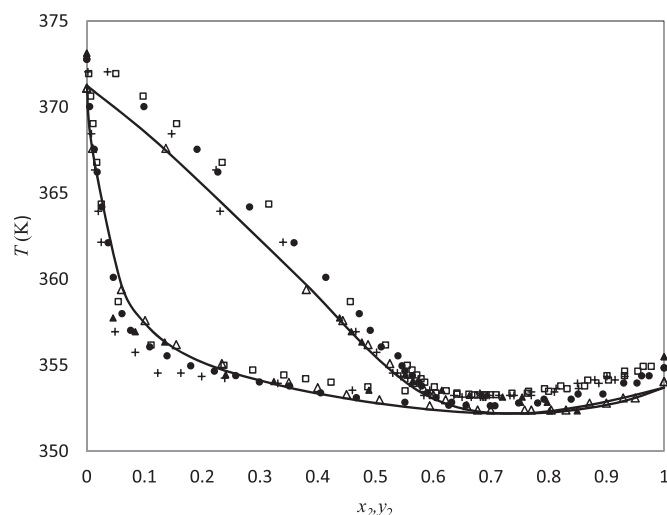


Fig. 4. Experimental VLE data of IPA (2) + Water (3) binary system. Solid line: $T_{\text{cal-}x_2}$, $T_{\text{cal-}y_2}$ (calculated from Wilson model coefficients) This work at 94.67 kPa; (Δ) $T_{\text{exp-}x_2}$, $T_{\text{exp-}y_2}$ at 94.67 kPa from this work; (□) Reference [5] at 101.32 kPa; (●) Reference [6] at 100 kPa; (▲) Reference [7] at 101.95–102.27 kPa; (+) Reference [8] at 101.3 kPa.

Table 8
Performance of optimized Wilson parameters for IPA (2) + Water (3) system at 94.67 kPa from the present work in predicting T and y of data sets from literature.

Reference	P (kPa)	rmsd T^a	rmsd y_2^b
[5]	101.32	1.0439	0.0282
[6]	30	1.9586	0.0455
	60	1.7348	0.0421
	100	1.5168	0.0415
[7]	101.95–102.27	2.3017	0.0329
[8]	101.3	2.0841	0.0399

^a $\sigma(T) = [\sum(T_{\text{exp}} - T_{\text{cal}})^2/n]^{0.5}$ K where n is the number of observations.

^b $\sigma(y_2) = [\sum(y_{2\text{exp}} - y_{2\text{cal}})^2/n]^{0.5}$ where n is the number of observations, y vapour phase mole fraction.

Table 9

Experimental VLE data of the quaternary system Methanol (1) + IPA (2) + Water (3) + Glycerol (4) at different pressures.

$P = 53.33$ kPa							
$^a x_1$	$^a x_2$	$^a x_3$	T (K)	$^b y_1$	$^b y_2$	$^b y_3$	$^b y_4 \cdot 10^{-5}$
0.1422	0.6808	0.0457	331.55	0.4026	0.5818	0.0157	0.2667
0.1313	0.5038	0.0739	333.70	0.3330	0.6462	0.0208	0.4092
0.0523	0.2905	0.3531	338.25	0.1731	0.6777	0.1494	0.4361
0.0993	0.0525	0.7500	339.75	0.3518	0.2905	0.3576	0.0795
0.5978	0.2151	0.1076	324.80	0.7038	0.2682	0.0280	0.0257
0.0563	0.4094	0.2850	337.65	0.1923	0.6819	0.1258	0.4091
0.5978	0.2151	0.1076	324.40	0.7038	0.2682	0.0280	0.0257
0.1214	0.5806	0.0911	333.85	0.3436	0.6257	0.0306	0.3474
0.1891	0.1250	0.6205	333.35	0.4611	0.3092	0.2297	0.0293
0.6100	0.2271	0.0966	324.45	0.7075	0.2670	0.0255	0.0214
0.1879	0.1988	0.2416	333.40	0.4140	0.5173	0.0686	0.2762
0.3191	0.0802	0.5565	330.80	0.6169	0.2048	0.1783	0.0141
0.0606	0.2332	0.2790	338.30	0.1833	0.7175	0.0993	0.5930
0.1093	0.4762	0.0940	334.20	0.2918	0.6807	0.0276	0.4749
0.2000	0.3285	0.4100	331.45	0.4576	0.3861	0.1563	0.0337
0.0576	0.0226	0.8887	344.80	0.2691	0.1894	0.5415	0.0364
$P = 66.66$ kPa							
$^a x_1$	$^a x_2$	$^a x_3$	T (K)	$^b y_1$	$^b y_2$	$^b y_3$	$^b y_4 \cdot 10^{-5}$
0.1422	0.6808	0.0457	335.95	0.4037	0.5842	0.0122	0.3648
0.1313	0.5038	0.0739	338.95	0.3319	0.6534	0.0147	0.5703
0.0993	0.0525	0.7500	345.05	0.3583	0.296	0.3457	0.0715
0.5978	0.2151	0.1076	329.40	0.7028	0.2711	0.0261	0.034
0.0563	0.4094	0.2850	342.70	0.1965	0.696	0.1075	0.4961
0.5978	0.2151	0.1076	329.70	0.7029	0.2711	0.0261	0.034
0.1214	0.5806	0.0911	337.70	0.3466	0.6298	0.0237	0.4654
0.1891	0.1250	0.6205	338.20	0.4658	0.3101	0.2242	0.0275
0.6100	0.2271	0.0966	329.75	0.7055	0.2705	0.024	0.0287
0.1879	0.1988	0.2416	338.60	0.4148	0.5319	0.0534	0.3475
0.3191	0.0802	0.5565	335.75	0.6238	0.2011	0.1751	0.0132
0.0606	0.2332	0.2790	343.10	0.1837	0.7402	0.0762	0.7516
0.1093	0.4762	0.0940	339.30	0.2919	0.6885	0.0196	0.6556
0.0576	0.0226	0.8887	350.10	0.2726	0.1886	0.5389	0.0289
$P = 79.99$ kPa							
$^a x_1$	$^a x_2$	$^a x_3$	T (K)	$^b y_1$	$^b y_2$	$^b y_3$	$^b y_4 \cdot 10^{-4}$
0.1422	0.6808	0.0457	339.90	0.3953	0.5930	0.0117	0.0450
0.1313	0.5038	0.0739	342.50	0.3194	0.6659	0.0146	0.0731
0.0993	0.0525	0.7500	350.10	0.3530	0.2911	0.3561	0.0099
0.5978	0.2151	0.1076	333.50	0.6973	0.2761	0.0268	0.0048
0.0563	0.4094	0.2850	347.45	0.1873	0.7046	0.1082	0.0644
0.5978	0.2151	0.1076	334.50	0.6973	0.2761	0.0268	0.0048
0.1214	0.5806	0.0911	342.35	0.3363	0.6404	0.0233	0.0589
0.1891	0.1250	0.6205	343.50	0.4573	0.3113	0.2314	0.0038
0.6100	0.2271	0.0966	333.75	0.6988	0.2766	0.0246	0.0041
0.1879	0.1988	0.2416	343.25	0.4065	0.5384	0.0552	0.0483
0.3191	0.0802	0.5565	340.50	0.6225	0.1960	0.1815	0.0019
0.0606	0.2332	0.2790	348.55	0.1753	0.7469	0.0779	0.1000
0.1093	0.4762	0.0940	342.30	0.2803	0.7001	0.0196	0.0843
0.0576	0.0226	0.8887	355.35	0.2675	0.1851	0.5475	0.0039
$P = 95.33$ kPa							
$^a x_1$	$^a x_2$	$^a x_3$	T (K)	$^b y_1$	$^b y_2$	$^b y_3$	$^b y_4 \cdot 10^{-4}$
0.1422	0.6808	0.0457	344.85	0.3772	0.6092	0.0136	0.0672
0.1313	0.5038	0.0739	347.05	0.2994	0.6823	0.0184	0.1093
0.0993	0.0525	0.7500	354.70	0.3315	0.2904	0.3781	0.0191
0.5978	0.2151	0.1076	338.80	0.6874	0.2831	0.0295	0.008
0.0563	0.4094	0.2850	351.50	0.1696	0.707	0.1234	0.103
0.5978	0.2151	0.1076	338.60	0.6875	0.2831	0.0295	0.008
0.1214	0.5806	0.0911	346.35	0.316	0.6565	0.0277	0.09
0.1891	0.1250	0.6205	348.40	0.4362	0.316	0.2478	0.0077
0.6100	0.2271	0.0966	338.60	0.6896	0.2836	0.0268	0.0067
0.1879	0.1988	0.2416	348.50	0.3784	0.5537	0.068	0.0773
0.3191	0.0802	0.5565	345.10	0.6061	0.1985	0.1956	0.0039
0.0606	0.2332	0.2790	353.05	0.1553	0.7499	0.0948	0.1517
0.1093	0.4762	0.0940	348.30	0.2603	0.7151	0.0246	0.1263
0.0576	0.0226	0.8887	359.70	0.2529	0.1846	0.5624	0.008

Standard uncertainties u are $u(P) = 0.17$ kPa, $u(T) = 0.01$ K, $u(x_i) = 10^{-5}$.

^a x liquid phase mole fraction.

^b y vapour phase mole fraction (calculated using Wilson model).

glycerol-rich region at 101.3 kPa. From Fig. 2, it can be observed that the experimental data of this work at 94.93 kPa is in good agreement with the experimental VLE data reported by Oliveira et al. [4] at 101.3 kPa. The effect of pressure on the VLE behaviour of IPA (2) + Glycerol (4) has been represented in Fig. 3. Experimental and calculated T - x_2 and calculated y_2 vs. T_{cal} and T_{exp} have been plotted in Fig. 3 for 66.7 and 94.93 kPa y_2 values have been calculated from the Wilson model using optimised binary interaction parameters obtained by correlation of experimental VLE data at the different pressure conditions studied.

For the IPA (2) + Water (3) binary system measured experimentally at 94.67 kPa. Both the activity coefficient models satisfactorily correlate the experimental VLE data of this system with the Wilson model giving a slightly better fit as can be observed from the rmsd T values presented in Table 7. Wilson model predicts an azeotrope for the IPA (2) + Water (3) binary system at $x_2 = 0.735$

and $T = 352.4$ K at 94.67 kPa. The T - x_2 - y_2 data for this system obtained from selected literature is compared with the experimental data reported in this work Fig. 4. Vapour phase mole fraction values for this work plotted in Fig. 4 have been obtained from Wilson model. The optimized parameters of the Wilson model obtained from this work for the IPA (2) + Water (3) at 94.67 kPa were applied to the data sets given in Refs. [21–24]. The deviations of the experimental equilibrium temperatures and vapour phase mole fractions of the low-boiling compound i.e. IPA from the values thus calculated are given in Table 8. The experimental and literature data are in good agreement. The optimized binary interaction parameters obtained for the IPA (2) + Water (3) system at 94.67 kPa in this work have been used to predict VLE of data sets from literature [21–24]. The rmsd in T and y_2 are presented in Table 8. The parameters from this work have predicted T and y_2 of literature data reasonably well.

Table 10
Parameters of Wilson and NRTL models used in quaternary predictions at different pressures.

Binary system	Methanol (1) + IPA (2) ^a	Methanol (1) + Water (3) ^b	Methanol (1) + Glycerol (4) ^b	IPA (2) + Water (3) ^c	IPA (2) + Glycerol (4) ^c	Water (3) + Glycerol (4) ^b
Quaternary pressure	53.33 kPa					
Wilson parameters ^d	$\sigma(T)^f = 0.326$					
A_{12}	0.3668	0.8957	0.9182	0.1362	−0.0252	1.2833
A_{21}	0.3659	0.5963	0.6829	0.802	−0.0144	1.6102
P/kPa	53.3	56.03	45.3	94.66	53.33	54.72
NRTL parameters ^e	$\sigma(T)^f = 10.05$					
A_{12}	1.0758	0.3363	0.283	0.6933	0.7893	−0.3431
A_{21}	1.0737	0.2947	0.1702	1.6785	0.9075	−0.3953
α	0.597	0.4	0.4	0.6806	−1.57	0.4
P/kPa	53.3	56.03	45.3	94.66	53.33	54.72
Quaternary pressure	66.67 kPa					
Wilson parameters ^d	$\sigma(T)^f = 0.327$					
A_{12}	0.3172	0.8851	0.9182	0.1362	−0.0071	1.6599
A_{21}	0.4109	0.544	0.6829	0.802	−0.0144	2.1342
P/kPa	66.7	67.38	45.3	94.66	66.7	63.84
NRTL parameters ^e	$\sigma(T)^f = 11.29$					
A_{12}	1.0398	0.3765	0.283	0.6933	0.7683	−0.7825
A_{21}	1.2634	0.3329	0.1702	1.6785	0.8549	−0.5076
α	0.6324	0.4	0.4	0.6806	−1.9217	0.4
P/kPa	66.7	67.38	45.3	94.66	66.7	63.84
Quaternary pressure	79.99 kPa					
Wilson parameters ^d	$\sigma(T)^f = 0.311$					
A_{12}	0.2144	0.8851	0.9182	0.1362	−0.0250	1.6599
A_{21}	0.5338	0.544	0.6829	0.802	−0.0150	2.1342
P/kPa	79.99	67.38	45.3	94.66	79.99	63.84
NRTL parameters ^e	$\sigma(T)^f = 10.96$					
A_{12}	0.8283	0.3765	0.283	0.6933	0.8017	−0.7825
A_{21}	1.4136	0.3329	0.1702	1.6785	0.9275	−0.5076
α	0.5806	0.4	0.4	0.6806	−1.4709	0.4
P/kPa	79.99	67.38	45.3	94.66	79.99	63.84
Quaternary pressure	95.2 kPa					
Wilson parameters ^d	$\sigma(T)^f = 0.297$					
A_{12}	0.1996	0.8941	0.9182	0.1362	−0.0230	1.2875
A_{21}	0.5948	0.6116	0.6829	0.802	−0.0095	1.9025
P/kPa	94.79	95.3	45.3	94.66	94.93	95.3
NRTL parameters ^e	$\sigma(T)^f = 10.92$					
A_{12}	1.0021	0.6326	0.283	0.6933	0.8124	−0.5989
A_{21}	1.665	0.024	0.1702	1.6785	0.9167	−0.3401
α	0.7097	0.4	0.4	0.6806	−1.4306	0.4
P/kPa	94.79	95.3	45.3	94.66	94.93	95.3

^a Reference [3].

^b Reference [1].

^c This work.

^d Wilson: $A_{12} = \Lambda_{12}$, $A_{21} = \Lambda_{21}$; $\Lambda_{ij} = (V_i^L/V_j^L) \exp(-(\lambda_{ij} - \lambda_{ji})/(RT))$.

^e NRTL: $A_{12} = \tau_{12}$, $A_{21} = \tau_{21}$; $\tau_{ij} = \Delta g_{ij}/(RT) = (g_{ij} - g_{ii})/(RT)$.

^f $\sigma(T) = [\sum (T_{exp} - T_{cal})^2/n]^{0.5}$ K, where n is the number of observations.

3.3. Isobaric quaternary VLE

The same experimental procedure as explained above was followed for studying the VLE behaviour of the quaternary system, Methanol (1) + IPA (2) + Water (3) + Glycerol (4). The quaternary system was studied at 53.33, 66.66, 79.99 and 95.2 kPa and the experimental data is presented in Table 9. The Wilson and NRTL activity coefficient model equations for multi-component are similar to the expressions for binary systems [22]. These equations have binary parameters and no ternary parameter term. An important aim of this phase equilibrium study is to validate predictions of multi-component VLE made using only binary interaction parameters by experiments.

The pressure equation for quaternary system evaluation is

$$P_{cal}/kPa = \sum (P_i^0/kPa)x_i\gamma_i \quad (4)$$

And the objective function used is the same as given by Eq. (3). MATLAB programmes using the chosen mathematical model and suitable multi-component equations of the thermodynamic model with pertinent pre-determined binary interaction parameters are used for prediction of multi-component VLE behaviour. From the present study, it can be observed that the IPA (2) + Glycerol (4) and IPA (2) + Water (3) binary systems are best represented by the Wilson model. Of the remaining constituent binary systems of this quaternary, Methanol (1) + Water (3), Methanol (1) + Glycerol (4) and Water (3) + Glycerol (4) are best represented by the Wilson activity coefficient model [1]. For the last constituent binary system, Methanol (1) + IPA (2), NRTL gave the best fit [3]. The quaternary experimental data was compared with predictions from Wilson and NRTL models calculated using only binary interaction parameters of the constituent systems. These are optimized parameters obtained by correlating experimental binary VLE data combining the selected thermodynamic models and suitable numerical methods using computer programming. Powell's Dog-Leg method is the numerical method used for the quaternary predictions as it was found to be superior to the Levenberg–Marquardt method in the binary correlation calculations. The parameters used for the quaternary analysis along with the rmsd in T obtained from the two models at the different pressures studied are presented in Table 10. Optimised parameters for the IPA (2) + Glycerol (4) and IPA (2) + Water (3) are from the present work while the parameters of the remaining binary systems are from our previous studies [1,3]. Wilson model satisfactorily predicts the phase behaviour of the multi-component. Vapour phase mole fractions calculated from Wilson model are presented in Table 9 along with experimental T - x data.

4. Conclusions

Experimental T - x_1 data of IPA (2) + Glycerol (4) binary system and Methanol (1) + IPA (2) + Water (3) + Glycerol (4) quaternary system were measured at atmospheric and sub-atmospheric pressures using a modified Sweitoslawsky ebulliometer while the binary system, IPA (2) + Water (3) was studied at the local atmospheric pressure. Experimental data of the binary systems was found to be consistent with literature data. Activity coefficient models Wilson, NRTL and UNIQUAC, group contribution method

UNIFAC and EoS models PR, SRK with WS mixing rules have been fitted to the experimental data of IPA (2) + Glycerol (4). Wilson and NRTL models were used to correlate the experimental data of the IPA (2) + Water (3) binary and predict VLE behaviour of the quaternary system. Wilson model represented the binary systems well and also gave the best fit to the multi-component system studied. Only binary interaction parameters of the constituent systems were used for multi-component predictions. Suitable numerical methods were employed and MATLAB programming was used for the correlation and prediction calculations. The optimized model parameters for the binary systems are reported. Multi-component VLE behaviour can be predicted with reasonable accuracy using only parameters estimated from binary experimental data.

Acknowledgements

The authors are grateful to CSIR file no. is 31/14(2547)/2013-EMRI (Council of Scientific & Industrial Research – India) for providing Research Associate fellowship to J.Soujanya.

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